

Stored Polarized Atomic Hydrogen for Møller Polarimetry

V.G. Luppov

University of Michigan Spin Physics Center

December 7, 2002

OUTLINE

1. Electron – spin – polarized atomic hydrogen stabilization

- 1.1 Principle
- 1.2 Michigan prototype apparatus

2. Target Thickness

- 2.1 Magnetic compression
- 2.2 Confinement time
- 2.3 Atomic hydrogen decay
- 2.4 Achievable density and thickness
- 2.5 Atomic hydrogen density monitoring

3. Target electron spin polarization

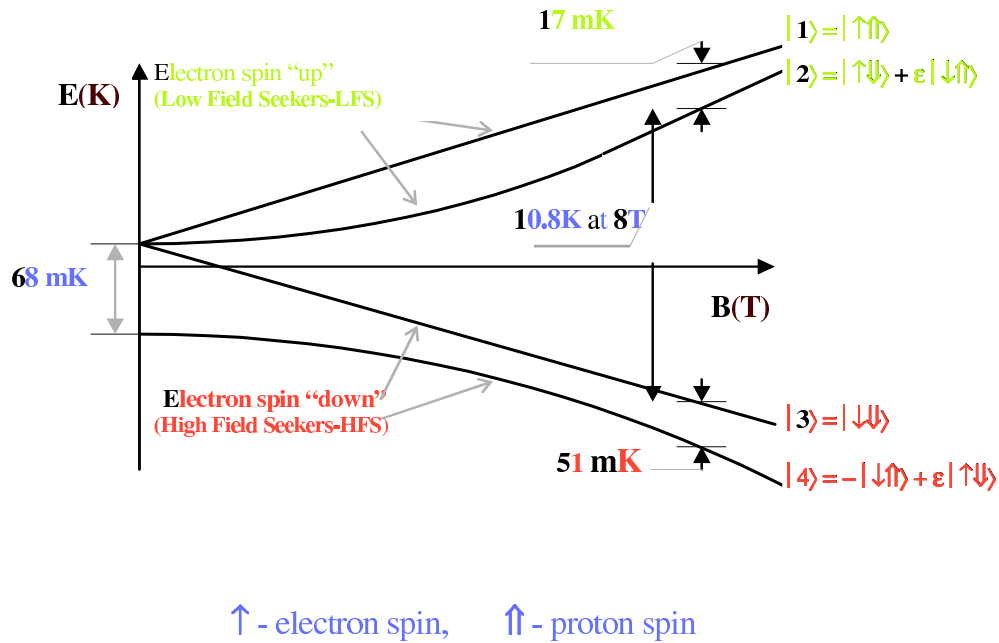
- 3.1 Polarization of incoming atomic hydrogen
- 3.2 Atoms with “wrong” spin admixture
- 3.3 Polarization “self-cleaning” mechanism
- 3.4 Depolarization processes
- 3.5 Polarization reversal

4. Residual gas backgrounds

- 4.1 Helium background
- 4.2 Accelerator residual gas background

5. Conclusions

Schematic Hyperfine Energy Level Diagram of Hydrogen Atom in Magnetic Field

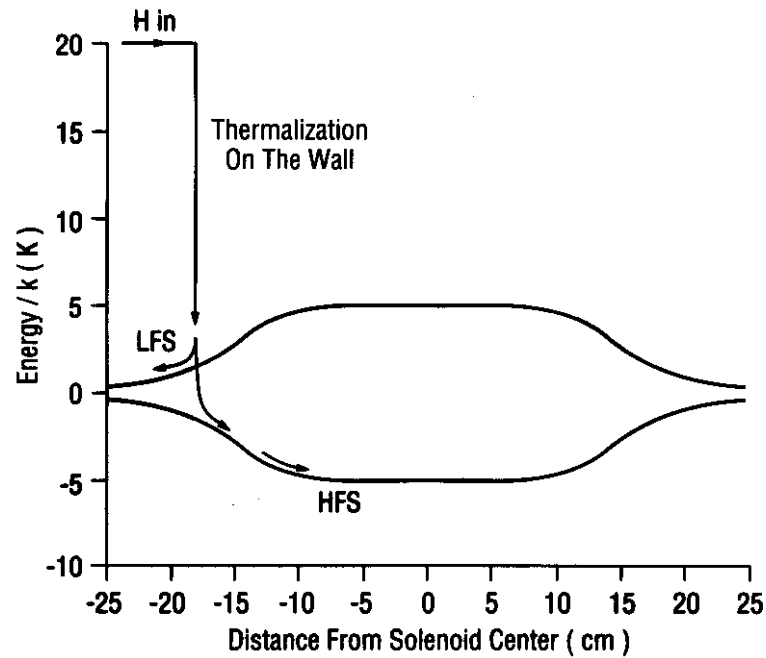


For instance, for a magnetic field of 8 T and the atomic hydrogen temperature of 300 mK, the ratio of electron spins - down to spins-up is $\exp(2\mu_B B/kT) = 3.6 \cdot 10^{15}$.

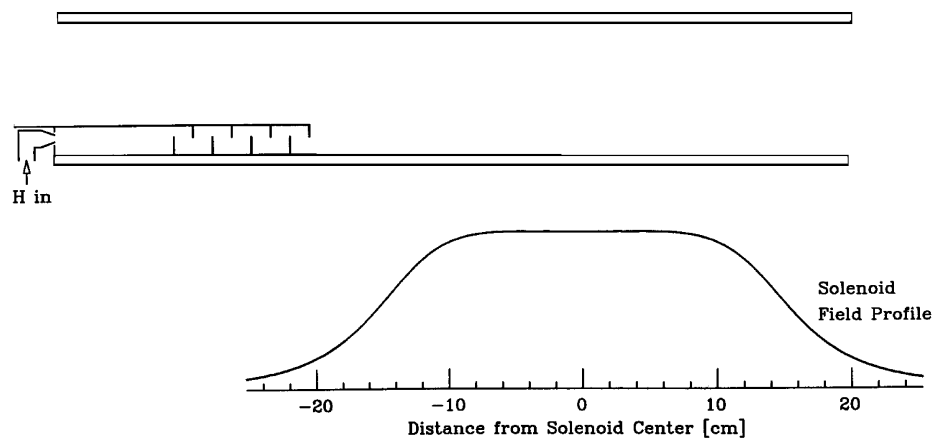
In order to stabilize atomic hydrogen several essential requirements have to be met:

- atoms must be made (by dissociating H_2);
- the electron spins must be polarized, so that atoms interact via the $^3\Sigma_u^+$ potential;
- the spin polarization must be maintained;
- the atomic hydrogen must be confined to a cell and the surface recombination must be suppressed.

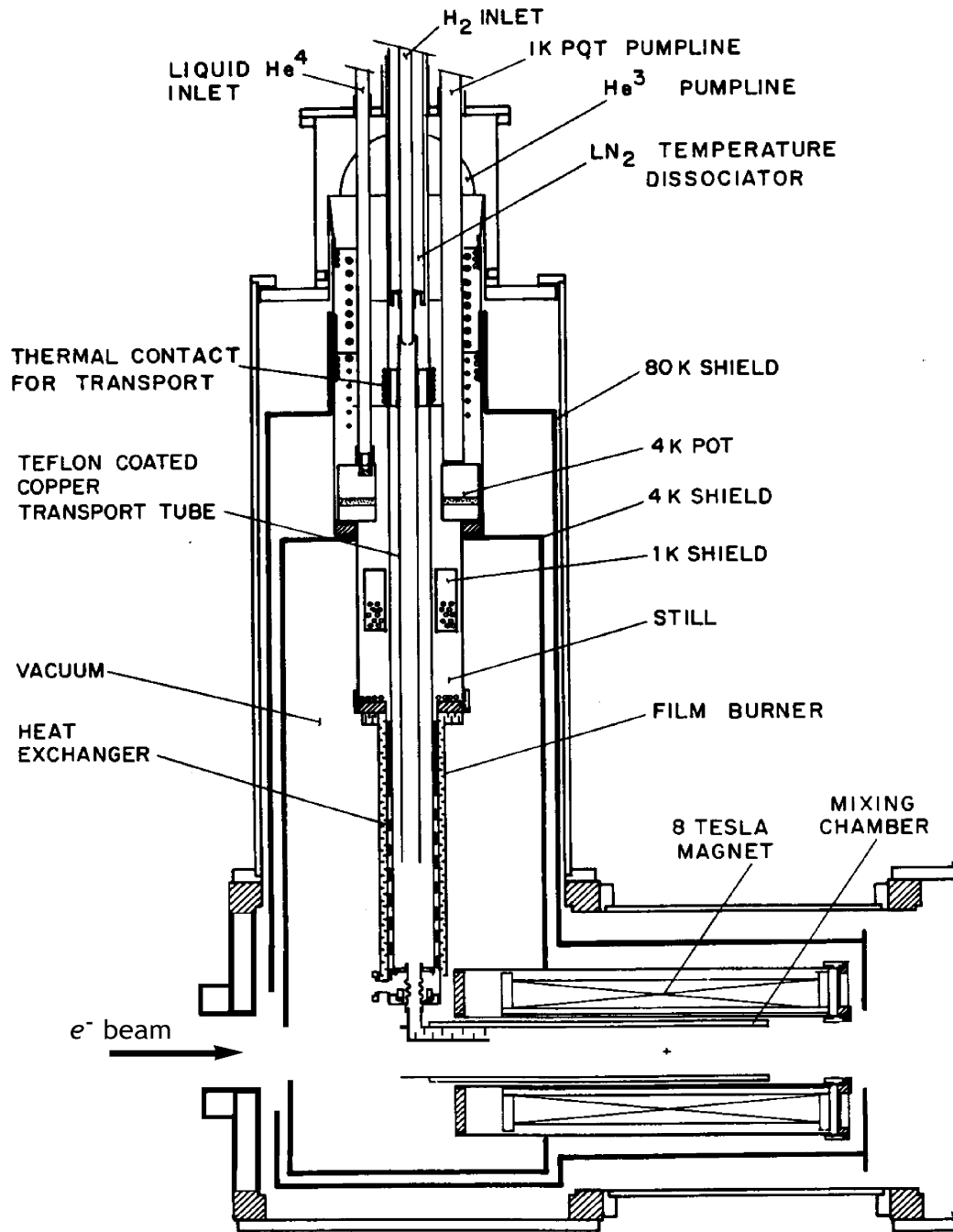
Potential Energy of Low Field Seekers ($|1\rangle$ and $|2\rangle$) and High Field Seekers ($|3\rangle$ and $|4\rangle$) Along the Solenoid Axis.



Storage Cell Displayed Relative to the Solenoid Field Profile



Schematic Diagram of the Michigan Prototype Jet Target



20 cm

The probability that the electron spin-up atoms enter the Stabilization Cell

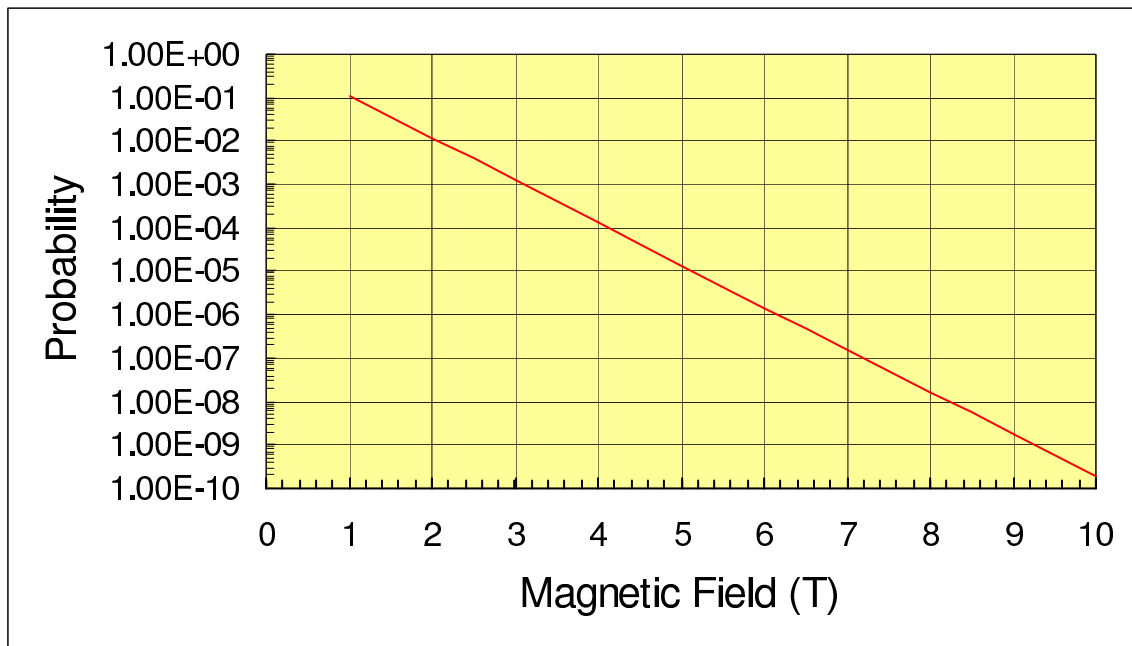
$$p = e^{-\mu B/kT}$$

where μ - Bohr magneton ($9.27 \cdot 10^{-24}$ J/T);

B - magnetic field;

K - Boltzmann constant ($1.38 \cdot 10^{-23}$ J/K)

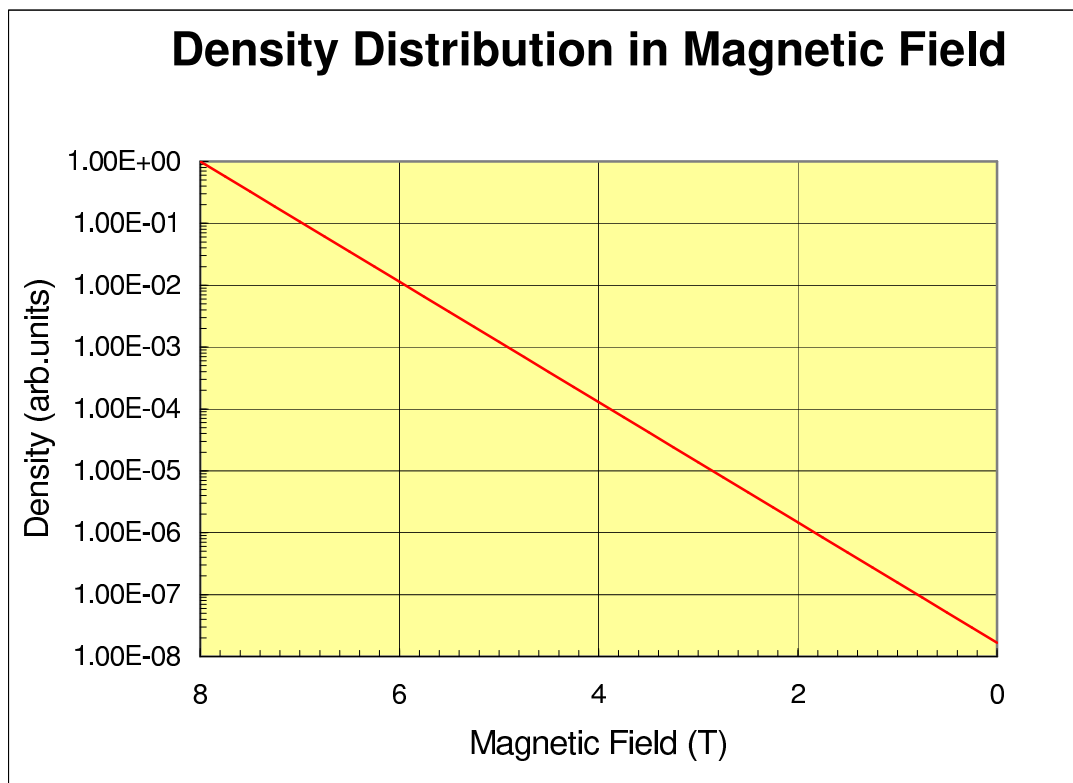
T - temperature ($T \equiv 300$ mK)



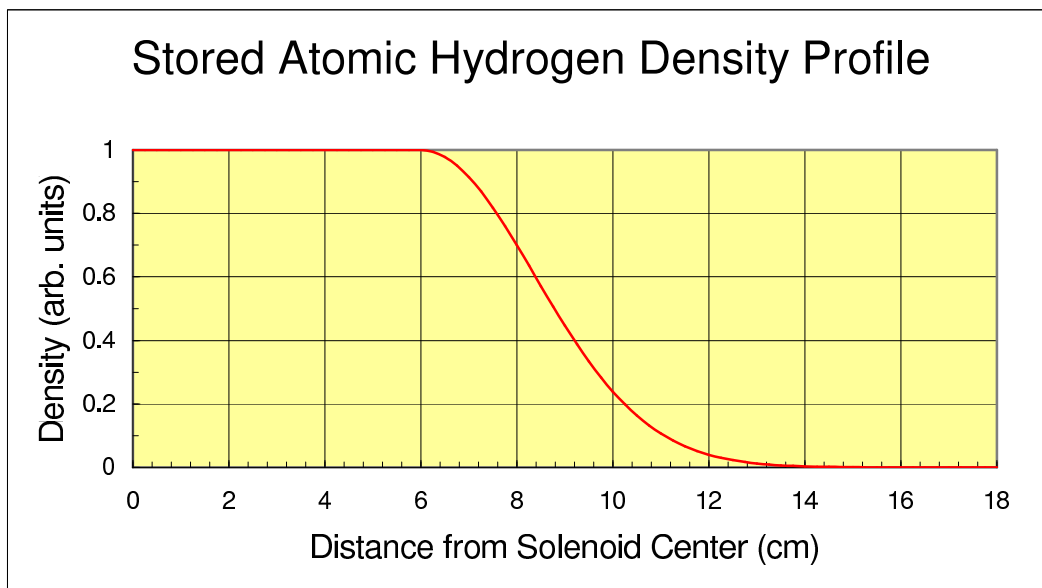
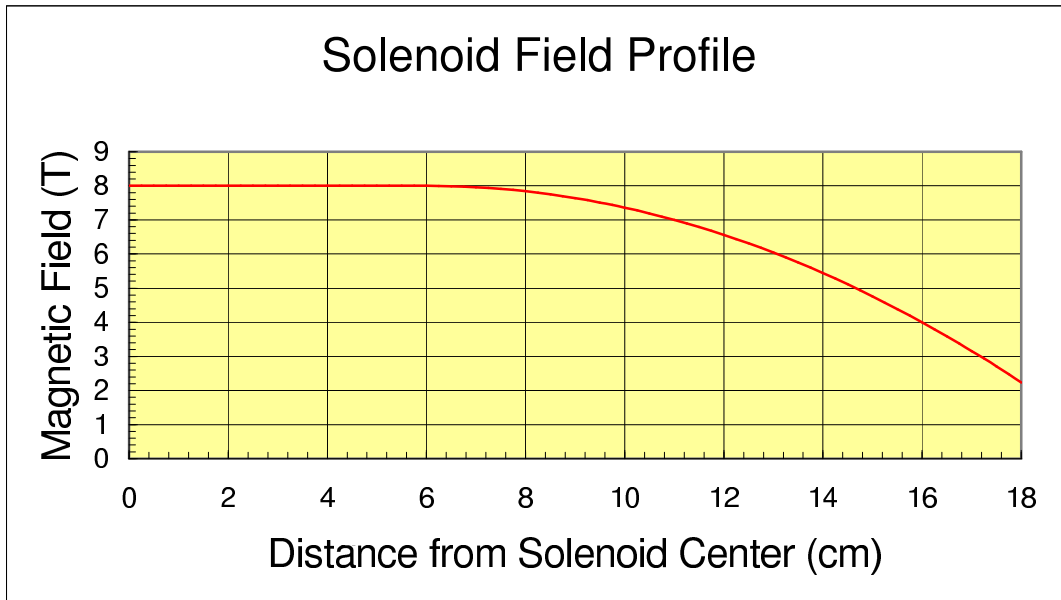
Magnetic Compression

Axial stored polarized atomic hydrogen density distribution:

$$n(z) = n(B_0)\exp[-\mu(B_0-B_z(z))/kT]$$



The axial longitudinal solenoid magnetic field decay can be approximated by $B_z(z) = B_0(1 - z^2/z_0^2)$ ($z_0=14$ cm):



Confinement Time

$$\frac{dn}{dt} = -n/\tau_{es}, \text{ where } \tau_{es} = \tau_{eso} e^{\mu B/kT}$$

$$\tau_{eso} = 4V_{eff}/Kv_{av}A \quad (\sim 50 \text{ ms})$$

V_{eff} - cell volume (for diameter $D=4$ cm, effective length

$L=19$ cm, $V_{eff}=239 \text{ cm}^3$)

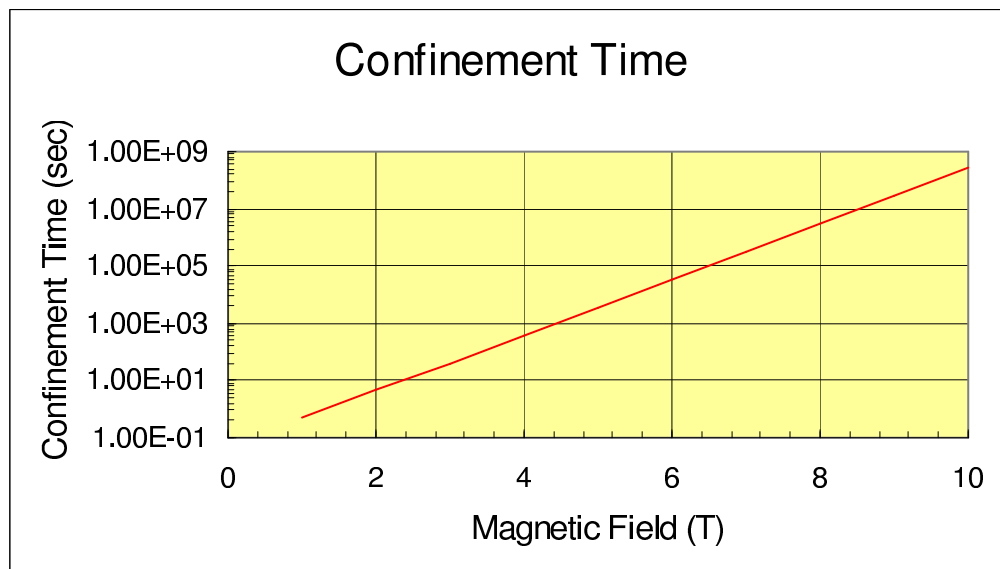
K - Clausing flow conductance fact

(Our cell $K=(15(L/D)+12(L/D)^2)/(20+38(L/D)+12(L/D)^2)$; $K=19$)

v_{av} - average atomic velocity

(for $T=300$ mk $v_{av}=(2/\sqrt{\pi})\sqrt{2kT/m}=80$ m/sec)

A - cell cross section (12.6 cm^2).



For 8 Tesla magnetic field $\tau_{es} = 3 \times 10^6$ sec: an ideal trap

Atomic Hydrogen Decay

Rate equations for hyperfine $|3\rangle$ and $|4\rangle$ states in case:

- there are no $|1\rangle$ and $|2\rangle$ states;
- low density ($< 10^{17}$ H/cm³) to ignore volume recombination
 $\{H + H + H \rightarrow H_2 + H\}$

$$dn_4/dt = \Phi_4/V_{\text{eff}} - 2 K_{44} n_4^2 - K_{43} n_4 n_3 - G_{43} (n_4 + n_3)(n_4 - n_3) - n_4/\tau_{\text{es}},$$

$$dn_3/dt = \Phi_3/V_{\text{eff}} - K_{43} n_4 n_3 + G_{43} (n_4 + n_3)(n_4 - n_3) - n_3/\tau_{\text{es}},$$

where n_i - atomic hydrogen $|i\rangle$ state density
 Φ_i - atomic hydrogen $|i\rangle$ state feed rate,
 V_{eff} - effective volume,
 K_{ij} - two-body recombination rate constants (occurring principally on the surface), $\{H + H + He \rightarrow H_2 + He\}$
 G_{43} - two-body magnetic dipole-dipole nuclear relaxation rate constant, $\{|3\rangle \rightarrow |4\rangle\}$
 τ_{es} - escaping time from the magnetic trap,

If $n_3 + n_4 \equiv n$ and $n_3 = n_4 = n/2$, then:

$$dn_4/dt = \Phi_4/V_{\text{eff}} - 2 K_{44} n_4^2 - K_{43} n_4 n_3 - G_{43} (n_4 + n_3)(n_4 - n_3) - n_4/\tau_{\text{es}},$$

+

$$dn_3/dt = \Phi_3/V_{\text{eff}} - K_{43} n_4 n_3 + G_{43} (n_4 + n_3)(n_4 - n_3) - n_3/\tau_{\text{es}}$$

$$dn/dt = \Phi/V_{\text{eff}} - K_s^{\text{eff}} n^2 - n/\tau_{\text{es}}$$

where $\Phi = \Phi_3 + \Phi_4$ - atomic hydrogen feed rate
 $K_s^{\text{eff}} = 1/2 (K_{43} + K_{44})$

Two-Body Recombination Rate Constants

$$K_{43} = 2.5 \cdot 10^{-8} (A/V) \Lambda^2 \exp(2\varepsilon_b/kT) T^{1/2} B^{-2}$$

$$K_{44} = 3 K_{43}$$

$$K_s^{\text{eff}} = 1/2 (K_{43} + K_{44}) = 5 \cdot 10^{-8} (A/V) \Lambda^2 \exp(2\varepsilon_a/kT) T^{1/2} B^{-2}$$

where A - storage cell surface area,

V - storage cell effective volume,

Λ - thermal deBroglie wavelength

$$(\Lambda = (h^2/2\pi k T M)^{1/2} = 1.74 \cdot 10^{-7} T^{-1/2})$$

ε_a - H adsorption energy on He surface ($\varepsilon_a \approx 1$ K)

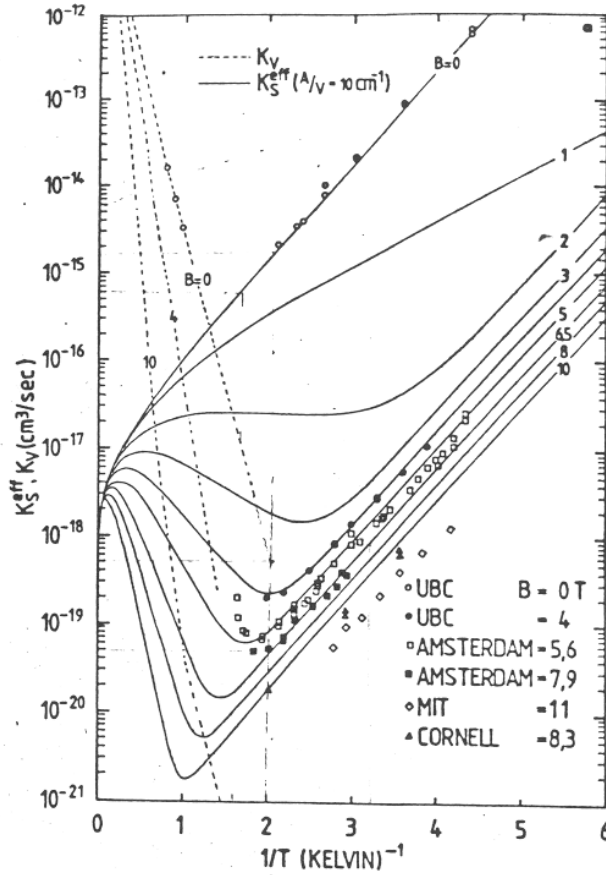
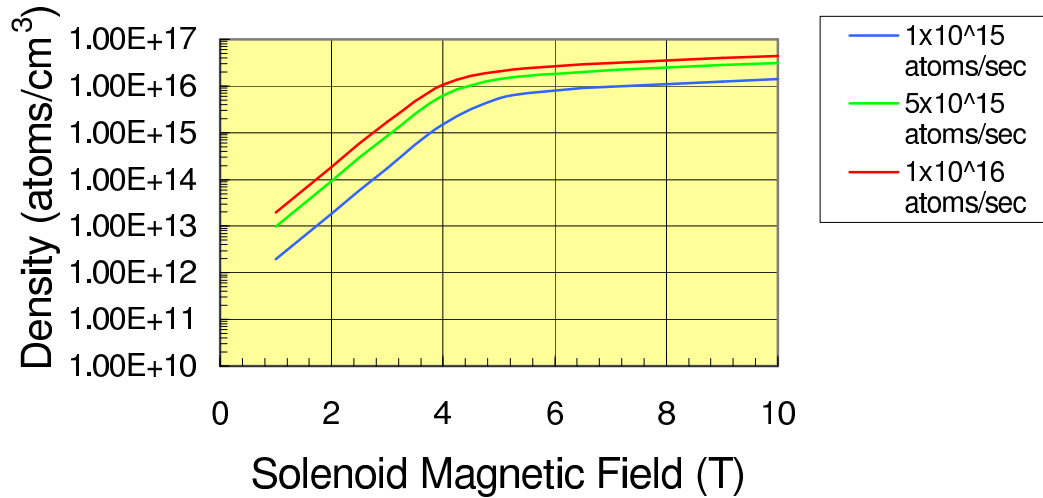
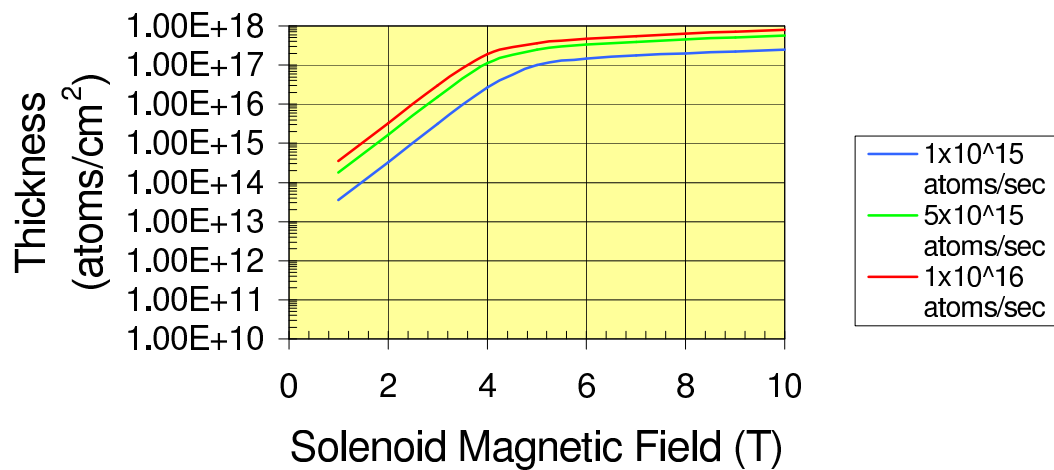


Fig. 5.9. (a) K_s^{eff} and K^* as a function of $1/T$ for H on ^4He surfaces. Solid line: K_s^{eff} ; dashed line: K^* . [UBC: $B = 0$ T, Hardy et al. (1980b), Morrow et al. (1981); $B = 4$ T, Statt et al. (1985); Amsterdam: $B = 7.9$ T, Matthey et al. (1981); $B = 5.6$ T, Matthey et al. (1984); MIT: Cline et al. (1981); Cornell: Yurke et al. (1983).]

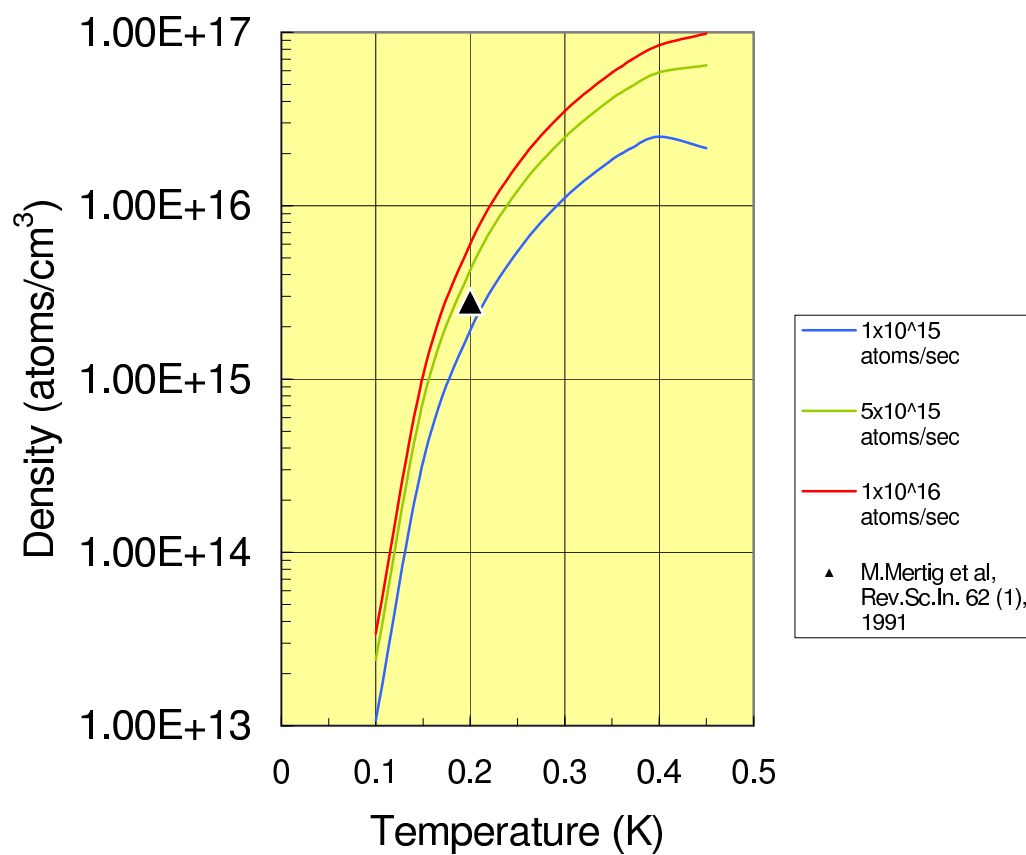
Stored Atomic Hydrogen **Density** vs Magnetic Field for Different H Feed Rates (T=0.3 K)



Stored Atomic Hydrogen **Thickness** vs Magnetic Field for Different H Feed Rates (T=0.3 K)



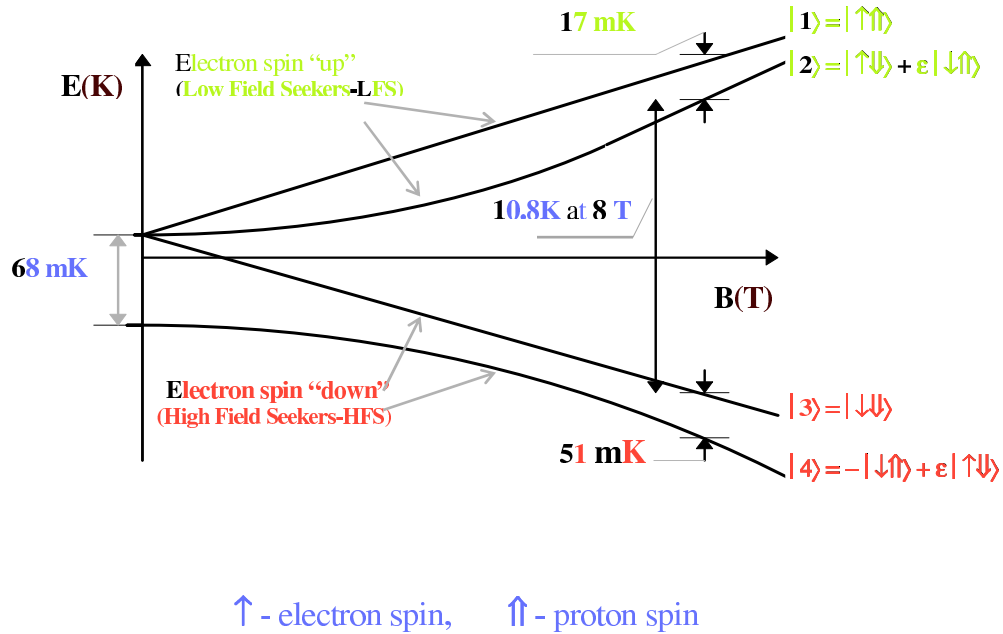
Stored Atomic Hydrogen Density vs Temperature (B =8 T)



Atomic Hydrogen Density Monitoring

Either a capacitive pressure gauge (Matthey, A.P.M., Walraven, J.T.M., and Silvera, I.F., *Phys. Rev.Lett.* **46**, 668 (1981),
or a bolometer monitor (Mertig, M., Luppov, V.G., Roser, T., and Vuaridel B., *Rev.Sci. Instrum.*, **62**(1), 251 (1991)) could be used for continues atomic hydrogen density measurements.

Target Electron-Spin- Polarization

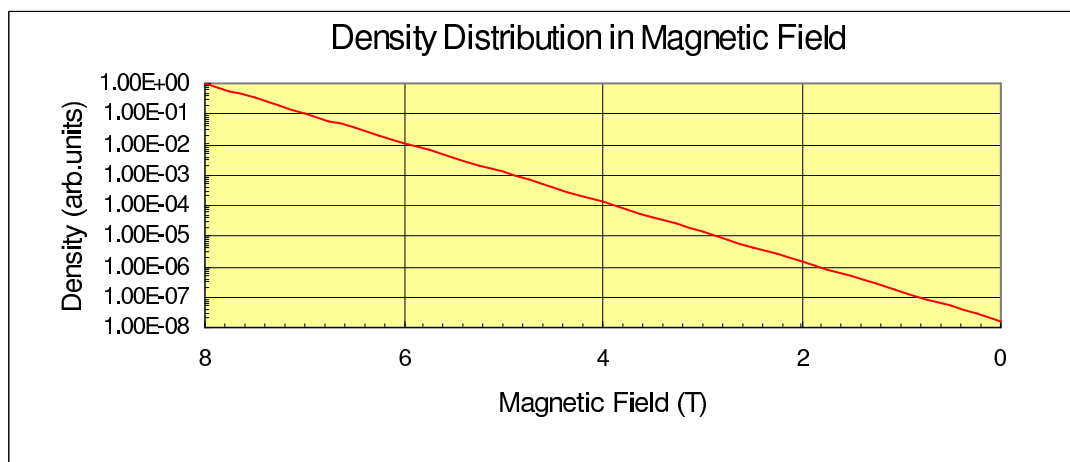
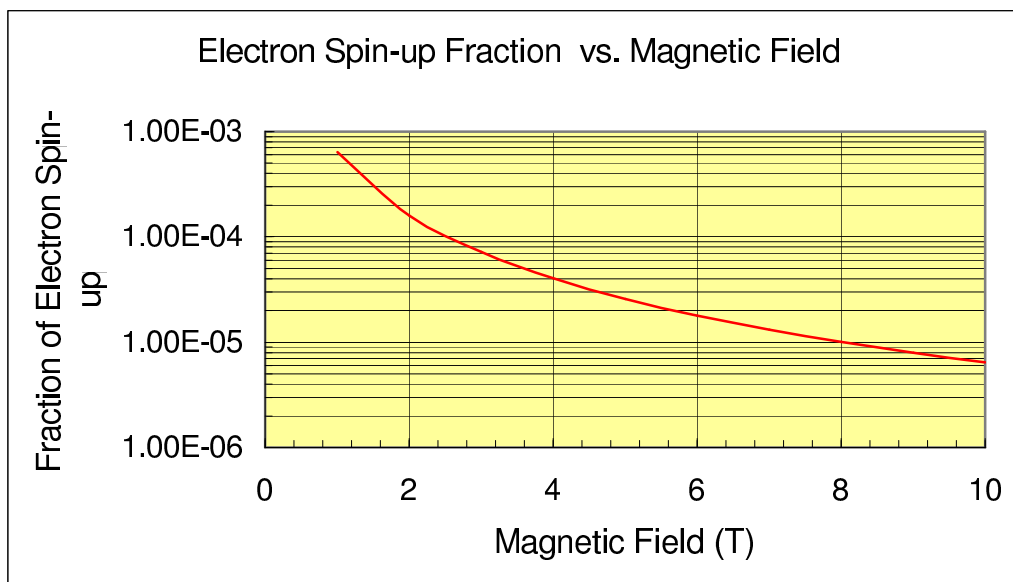


$\epsilon = \tan \theta$, where $\theta = 1/2 \cdot \arctan(a/[h(\gamma_e + \gamma_p)B])$

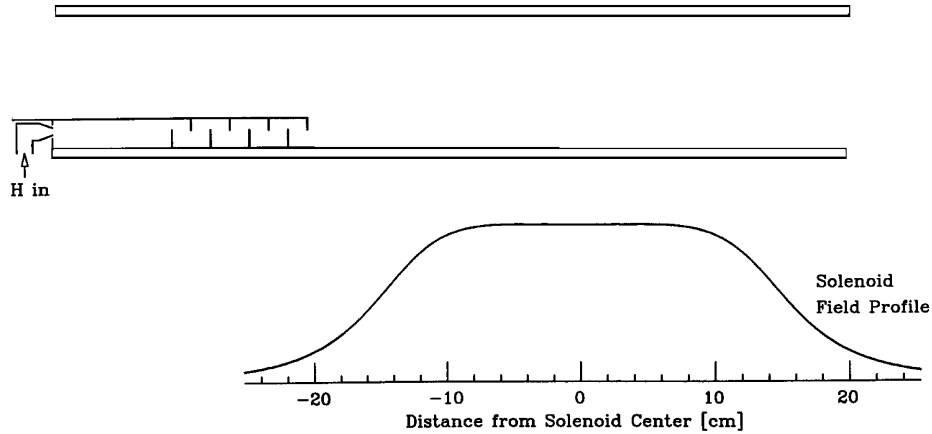
- $a = 9.42 \cdot 10^{-25}$ [J] - hyperfine coupling constant,
- $\gamma_e = 2.80 \cdot 10^{10}$ [T⁻¹s⁻¹] – electron gyromagnetic ratio,
- $\gamma_p = 4.26 \cdot 10^7$ [T⁻¹s⁻¹] – proton gyromagnetic ratio.

For the mixed state $|4\rangle$ a fraction of atoms with “wrong” electron–spin–up direction is $\tan^2 \theta$.

Target Electron-Spin- Polarization



Polarization “Self-Cleaning” Mechanism



$$\tau_{\text{eso}} = 4V_{\text{eff}}/Kv_{\text{av}}A \quad (\sim 50 \text{ ms})$$

V_{eff} - cell volume (for diameter $D=4$ cm, effective length $L=19$ cm, $V_{\text{eff}}=239 \text{ cm}^3$)

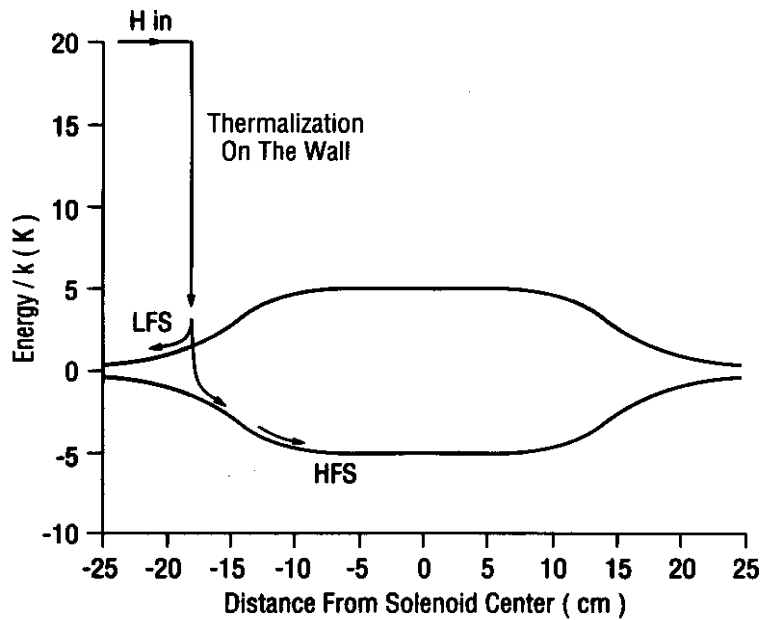
K - Clausing flow conductance fact

(Our cell $K=(15(L/D)+12(L/D)^2)/(20+38(L/D)+12(L/D)^2)$; $K=19$)

v_{av} - average atomic velocity

(for $T=300 \text{ mK}$ $v_{\text{av}}=(2/\sqrt{\pi})\sqrt{2kT/m}=80 \text{ m/sec}$)

A - cell cross section (12.6 cm^2).



Depolarization Processes

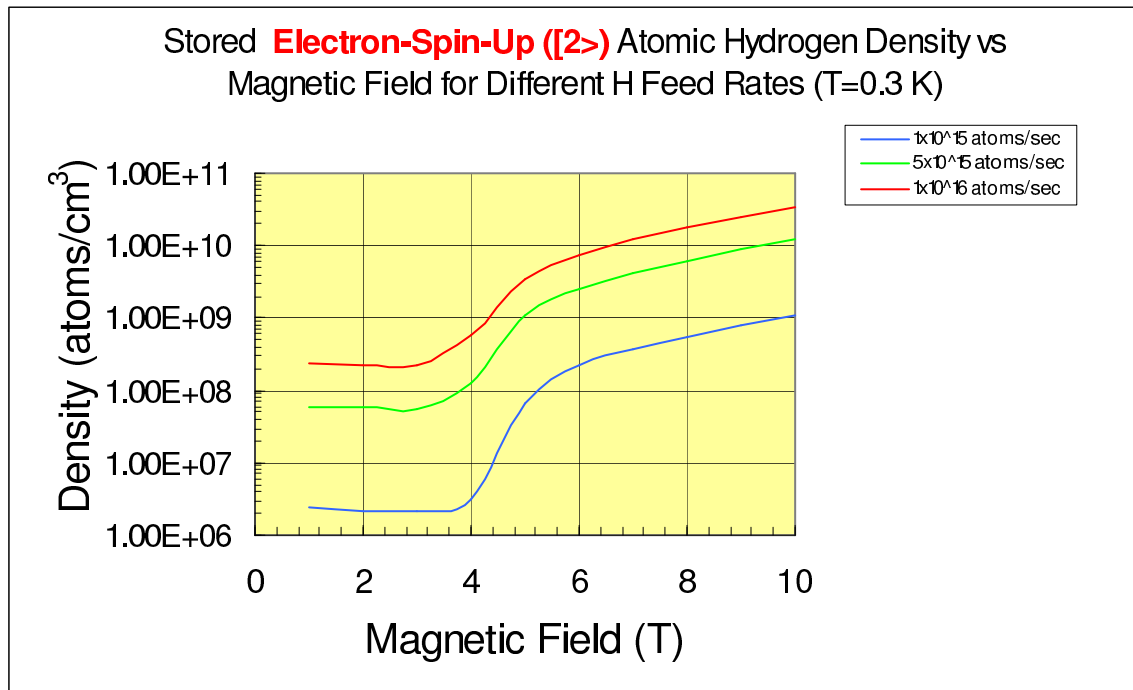
$$dn_2/dt = \xi K_{333} n_3^3 + G_{32} n_3^2 - n_2/\tau_{es0},$$

where $\xi = 0.91$ – a fraction of $\{ |3\rangle + |3\rangle + |3\rangle \rightarrow H_2 + |2\rangle \}$
 $(1 - \xi) \rightarrow \{ |3\rangle + |3\rangle + |3\rangle \rightarrow H_2 + |3\rangle \}$

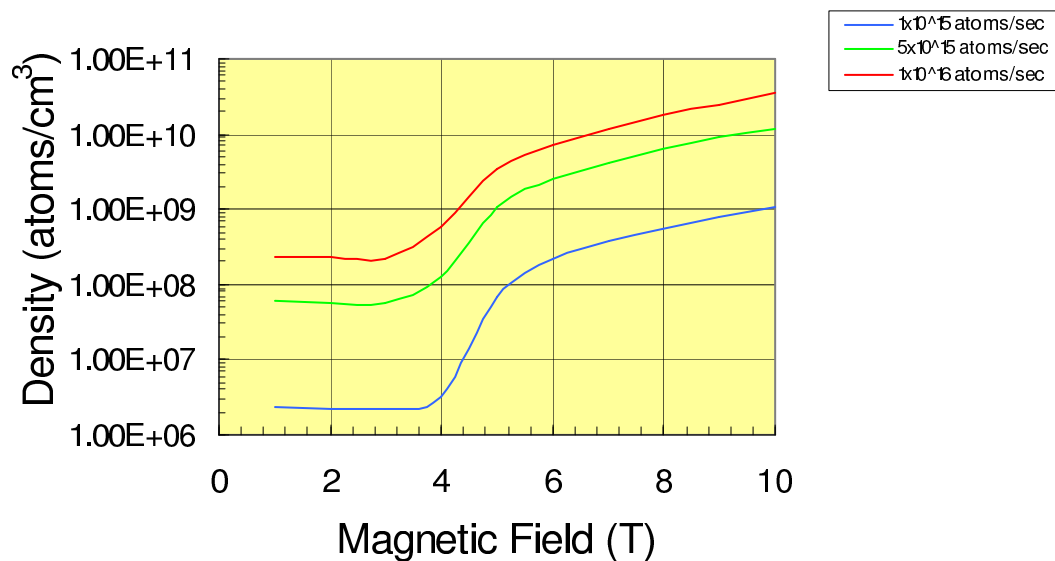
K_{333} - three-body recombination rate constant
 $(K_{333} \approx 9 \cdot 10^{-39} \text{ cm}^6/\text{sec})$,

G_{32} – electronic relaxation rate constant
 $(G_{32} = 1.1 \cdot 10^{-15} \exp(-1.35B/T) \text{ cm}^3/\text{sec})$

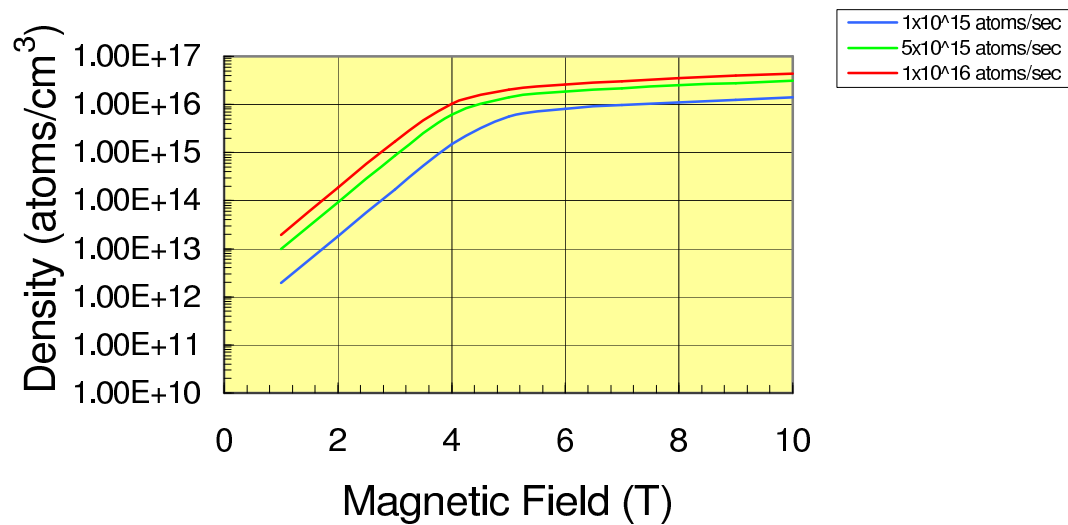
$\tau_{es0} = 0.05 \text{ sec}$



Stored **Electron-Spin-Up (I_2)** Atomic Hydrogen Density vs
Magnetic Field for Different H Feed Rates ($T=0.3$ K)

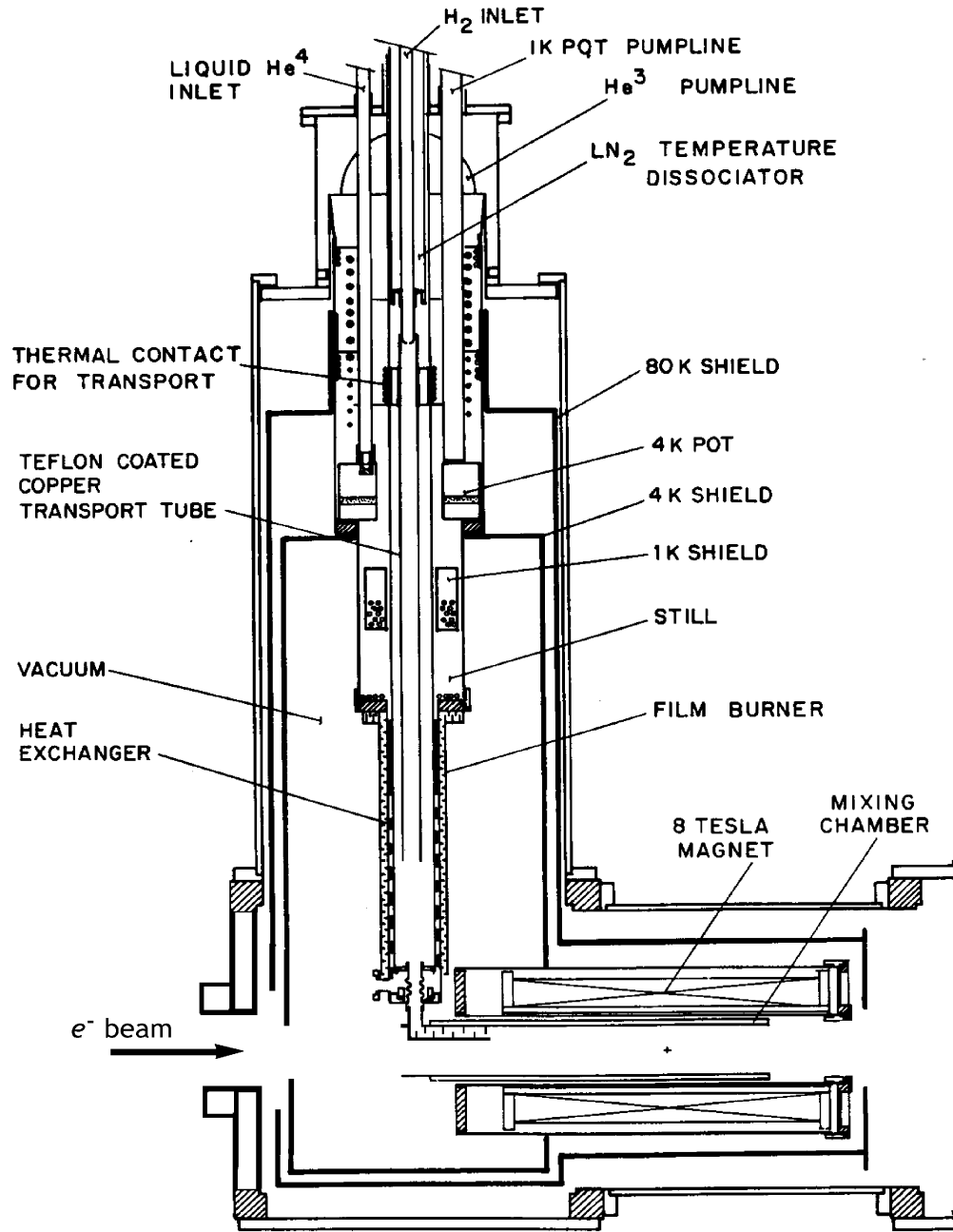


Stored Atomic Hydrogen Density vs Magnetic
Field for Different H Feed Rates ($T=0.3$ K)



Polarization Reversal

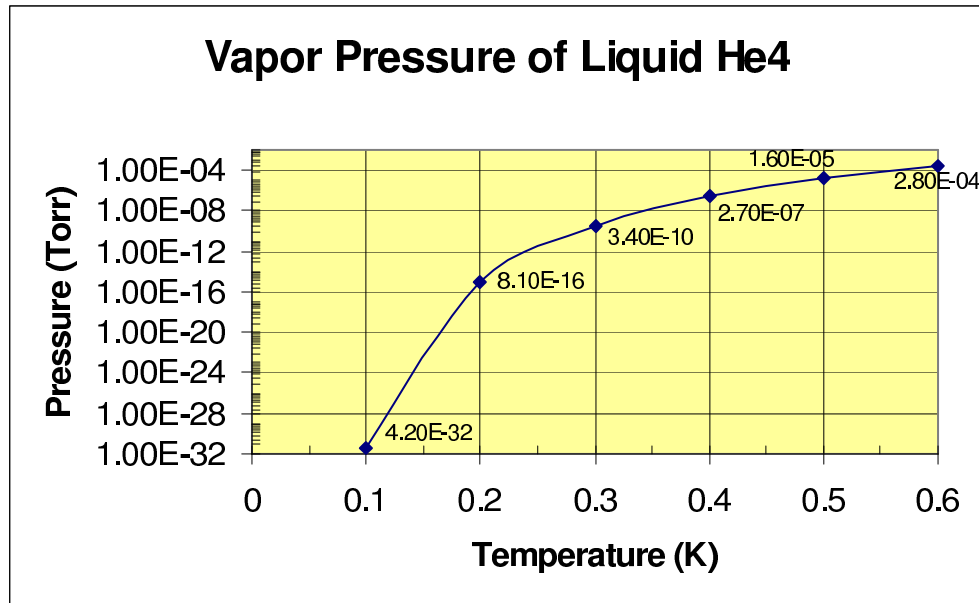
- To reverse the stored atomic hydrogen longitudinal polarization the magnetic field orientation has to be changed.
- For this purpose, the solenoid magnetic field should be first turned off, and then turned back on with the opposite polarity.
- Then the storage cell should be refilled with atomic hydrogen.
- The total procedure should not take longer than 1.5 hours.



Schematic Diagram of the Michigan Prototype Jet Target

Residual Gas Backgrounds

• Helium Background



- The superfluid film flows from the coldest towards the warmest surfaces until it evaporates.
- A special “film-burner” technique can be used to keep pressure, measured at 300 K, as low as $8 \cdot 10^{-7}$ Torr.
- This pressure corresponds to helium density of $n_{300K} = P/kT = 2.4 \cdot 10^{10}$ atoms/cm³ at 300 K, or $n_{0.3K} = n_{300K} \sqrt{300K/0.3K} = 8 \cdot 10^{11}$ atoms/cm³ at 300 mK, or $1.6 \cdot 10^{12}$ electrons/cm³, which is about 0.02% of the atomic hydrogen electron density (10^{16} electrons/cm³).
- This helium pressure could be significantly decreased by means of differential pumping.

• Accelerator Residual Gas Background (by EAC)

- At JLab, the accelerator residual gas pressure at room temperature is about $1 \cdot 10^{-5}$ Torr. It mainly consists of N₂ and water.
- The corresponding electron density is $3 \cdot 10^{12}$ electrons/cm³, or about 0.03% of the electron density of the atomic hydrogen.
- It should be mentioned that the residual gas pressure near the target would be significantly lower due to cryopumping.

Conclusions

- The considerations above show that stored, longitudinally electron-spin-polarized atomic hydrogen can be used as a pure, 100% polarized gas target.
- A thickness of at least $1 \cdot 10^{17}$ H/cm² can be reached with a target diameter of 4 cm and a length of 18 cm along the beam.

Possible Problems with the Electron Beam

The following possible problems have not yet been evaluated:

- Atomic hydrogen ionization and excitation by the beam.
- Heating of the copper storage cell by the beam 500 MHz RF.